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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CHEMISTRY **5070/01**

Paper 1 Multiple Choice **October/November 2008**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

3932853705

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

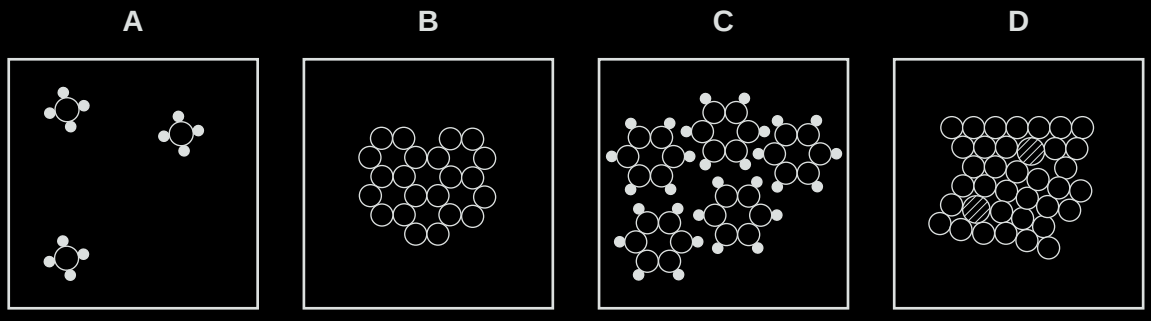
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
A copy of the Periodic Table is printed on page 16.

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8 Which gas diffuses at the same rate as nitrogen gas?

A carbon dioxide

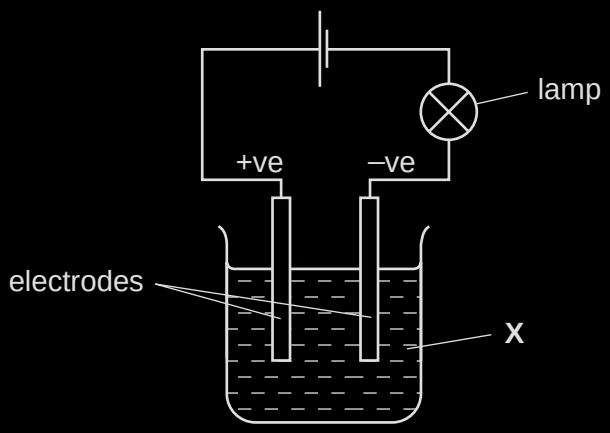
B carbon monoxide

C neon

D sulphur dioxide

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What is **X**?

A aqueous sodium chloride

B bromine

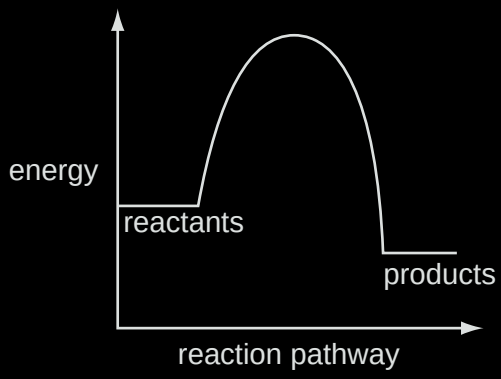
C molten sodium chloride

D mercury

16 What are the products formed at the electrodes during the electrolysis of molten magnesium chloride between carbon electrodes?

	positive electrode	negative electrode
A	oxygen	magnesium
B	magnesium	chlorine
C	chlorine	magnesium
D	chlorine	hydrogen

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Which information correctly describes the effect of the catalyst on the activation energy and enthalpy change for the reaction?

	activation energy	enthalpy change
A	decrease	decrease
B	increase	no change
C	increase	increase
D	decrease	no change

19 The fertiliser ammonium nitrate (NH_4NO_3, $M_r = 80$) is manufactured from ammonia (NH_3, $M_r = 17$) by a two-stage process.

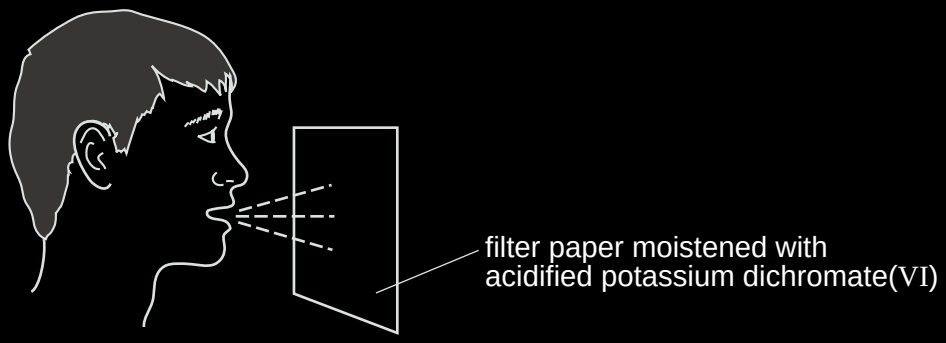
Stage 1 $NH_3 + 2O_2 \rightarrow HNO_3 + H_2O$

Stage 2 $HNO_3 + NH_3 \rightarrow NH_4NO_3$

What is the maximum mass of fertiliser that can be made if only 17 tonnes of ammonia is available?

A 34 tonnes **B** 40 tonnes **C** 80 tonnes **D** 97 tonnes

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A colour change from orange to green is observed if ethanol is present.

This shows that ethanol is

A an alkali.

B an indicator.

C an oxidising agent.

D a reducing agent.

21 In the Haber process, nitrogen and hydrogen react to form ammonia.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92\,kJ$$

Which factor increases **both** the speed of reaction **and** the amount of ammonia produced?

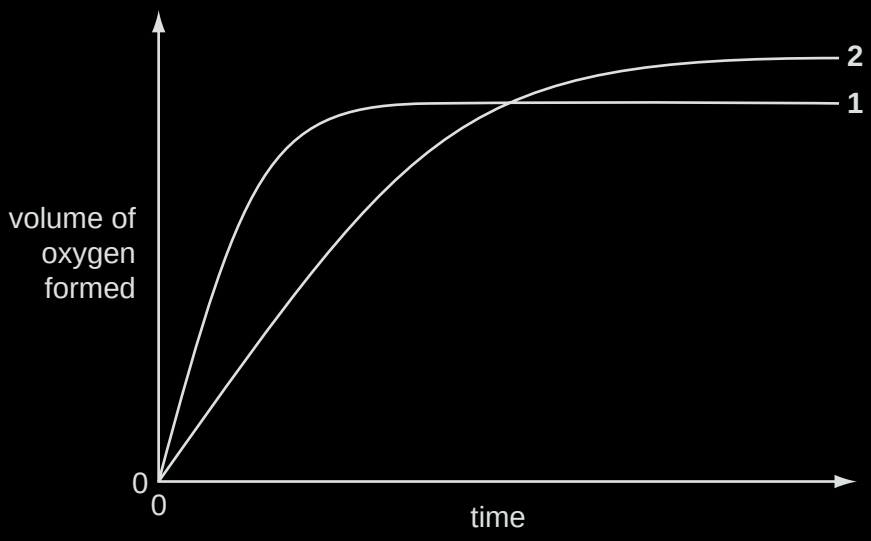
A addition of a catalyst

B decreasing the temperature

C increasing the pressure

D increasing the temperature

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Which alteration to the original experimental conditions would produce curve **2**?

A lowering the temperature

B adding some 0.1 mol/dm^3 hydrogen peroxide solution

C using less manganese(IV) oxide

D using a different catalyst

23 In which reaction is sulphur dioxide acting as an oxidising agent?

A $SO_2 + 2H_2O + Cl_2 \rightarrow H_2SO_4 + 2HCl$

B $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$

C $2SO_2 + O_2 \rightarrow 2SO_3$

D $SO_2 + 2H_2S \rightarrow 2H_2O + 3S$

24 Which element will burn in oxygen to form an acidic oxide?

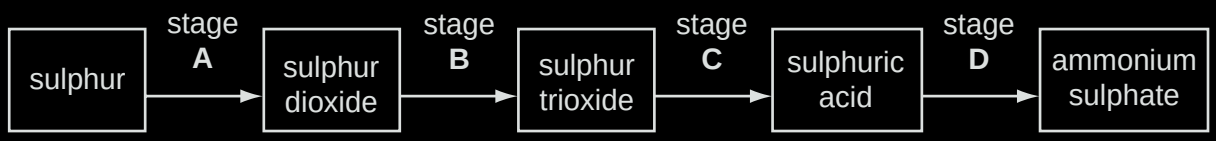
A calcium

B carbon

C iron

D magnesium

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28 One mole of compound **X** gives three moles of ions in aqueous solution. **X** reacts with ammonium carbonate to give an acidic gas.

What is compound **X**?

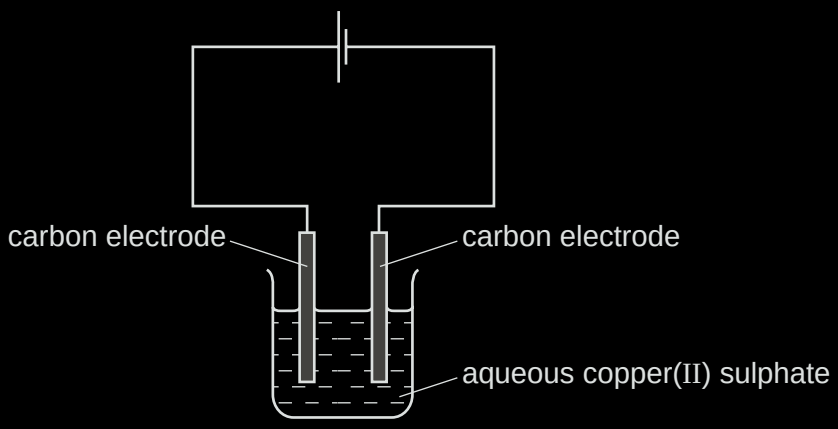
A calcium hydroxide

B ethanoic acid

C sodium hydroxide

D sulphuric acid

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Which ionic equations show the reactions at the electrodes?

1 $Cu^{2+} + 2e^- \rightarrow Cu$

2 $Cu \rightarrow Cu^{2+} + 2e^-$

3 $2H^+ + 2e^- \rightarrow H_2$

4 $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

A 1 and 2 only **B** 1 and 4 only **C** 2 and 3 only **D** 3 and 4 only

31 The element chromium liberates hydrogen from dilute hydrochloric acid although it does not react with cold water. When a piece of chromium is placed in lead(II) nitrate solution, crystals of lead appear.

What is the order of **decreasing** reactivity of the metals lead, calcium and chromium?

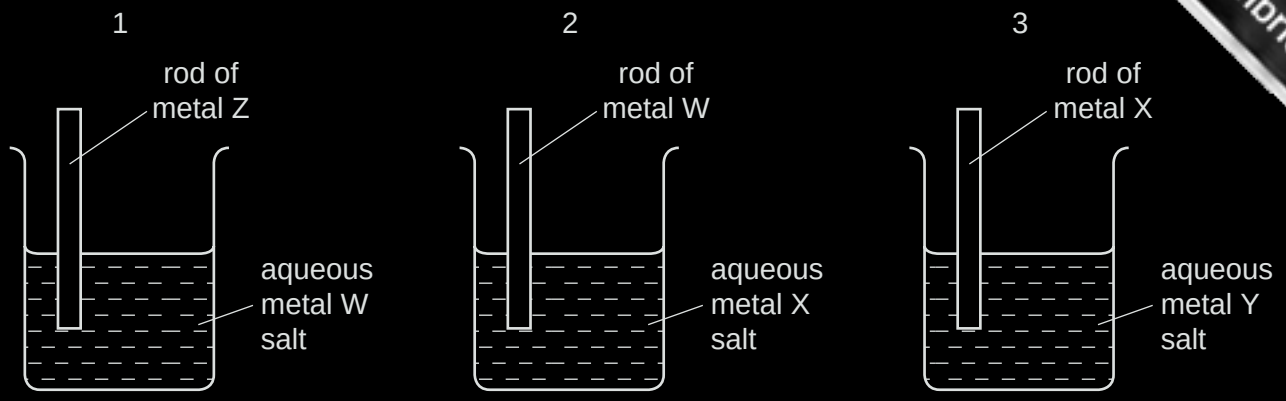
A calcium, chromium, lead

B calcium, lead, chromium

C chromium, calcium, lead

D lead, chromium, calcium

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In beaker 1 metal W is displaced from solution.
In beaker 2 metal X is displaced from solution.
In beaker 3 metal Y is displaced from solution.

What is the order of **decreasing** reactivity of the four metals?

	most reactive ⟶			least reactive
A	W	X	Y	Z
B	Z	W	X	Y
C	Z	X	W	Y
D	X	Y	W	Z

33 What is the function of silica, SiO_2, in the equation shown below?

$$CaO + SiO_2 \rightarrow CaSiO_3$$

A a basic oxide

B a reducing agent

C an acidic oxide

D an oxidising agent

34 Alloys are usually harder than the metals from which they are made.

Which difference between the metals explains the greater hardness of alloys?

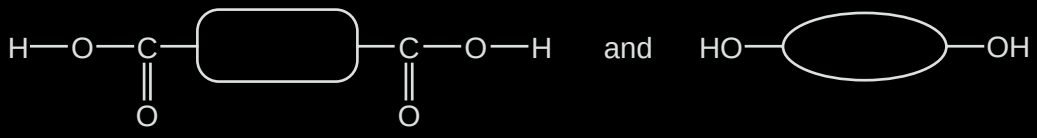
A atomic radius

B boiling point

C density

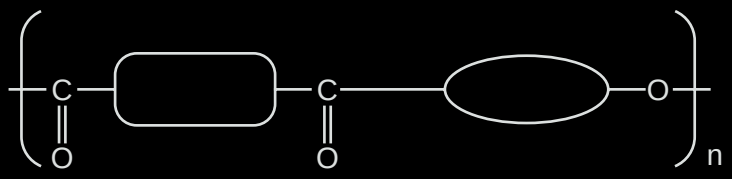
D malleability

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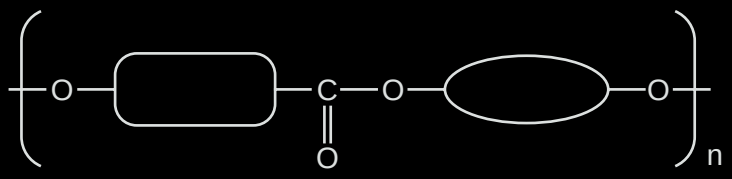


What is the repeat unit of the polymer?

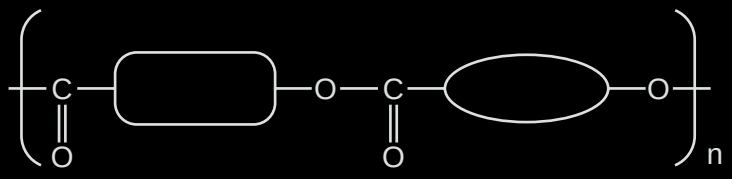
A



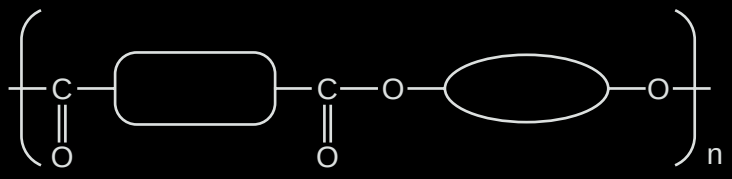
B



C



D



37 Which molecule does **not** undergo an addition reaction with alkenes?

A ammonia, NH_3

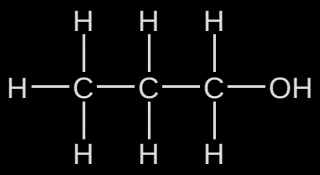
B bromine, Br_2

C hydrogen, H_2

D steam, H_2O

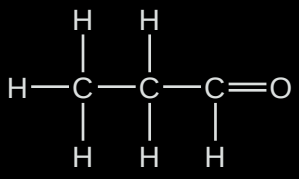
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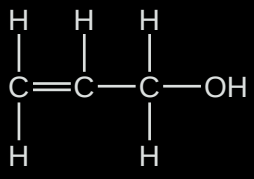


Which of the following is an isomer of propan-1-ol?

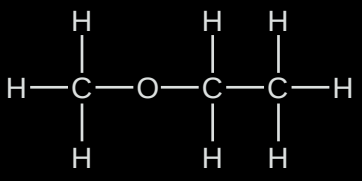
A



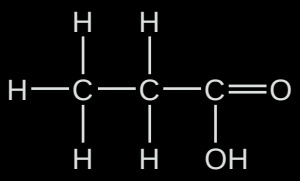
B



C



D



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DATA SHEET
The Periodic Table of the Elements

Group																	
I	II										III	IV	V	VI	VII	0
							1 H Hydrogen 1									4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89 †															

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92	Np Neptunium 93	Pu Plutonium 94	Am Americium 95	Cm Curium 96	Bk Berkelium 97	Cf Californium 98	Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101	No Nobelium 102	Lr Lawrencium 103

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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